

# Comp. Architecture

## Chap-1: Comp Abstraction & Technology

python  
hardware

(high lvl lang)

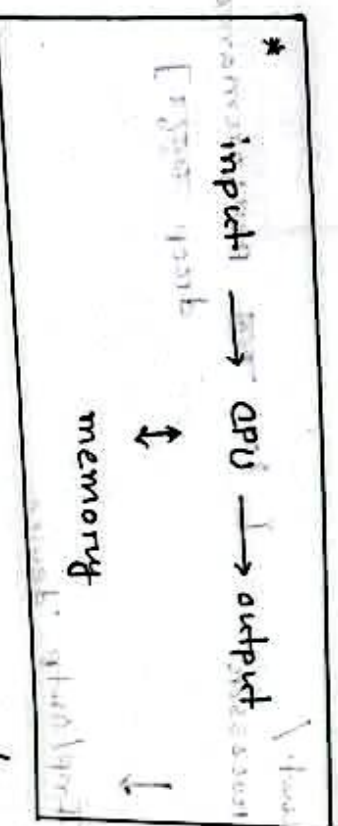
High lvl lang  $\xrightarrow{\text{convert}}$  low lvl lang (assembly)  $\rightarrow$

M.C (binary)  $\rightarrow$  circuit

| A | B | F |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

$\rightarrow$  AND Gate

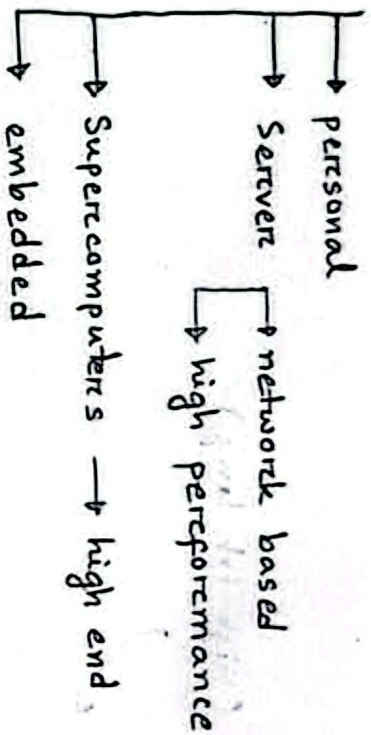
now, circuit optimizing



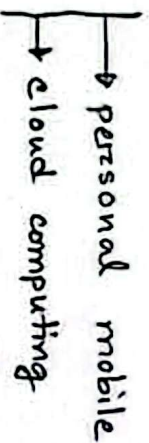
$\hookrightarrow$  von neuman architecture

# # Computer technology → Moore's law.

## Types



## The Post PC era -



## Understanding performance -

Algorithm  $O(n^2)$   
 $O(n \log n)$



circuit /  
processor



Imp/Outp device

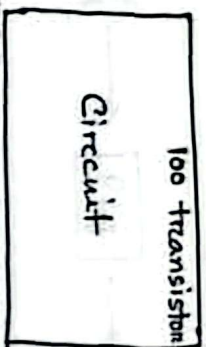
[ low ~~per~~ performance  
drop ~~per~~ ]

## Eight great Idea.

↳ Design for Moore's law :

Suppose, this time for a

↳ 2 yrs

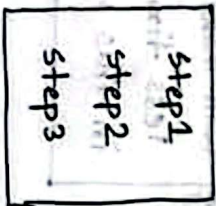


sol<sup>n</sup>

Moore's law: every year circuit a transistor 2 गुना बढ़े पाएगा।

↳ Use abstraction to simplify design:

task

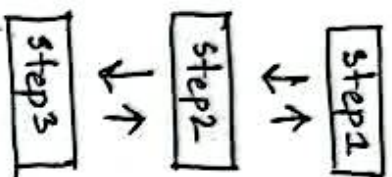


sol<sup>n</sup>



but → प्रोत्त step वास्तविक में it's failed & can't be changed.

So, by abstraction,



→ આમાના આમાના ઘાટ step  
 એ ઘાટ ઘાટ ઘાટ 1, 2, 3

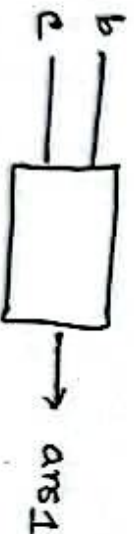
↳ Make the common case fast:



if we can convert anything & make it faster.

↳ performance via parallelism:

$$a = \underbrace{(b+c)}_{ans1} + \underbrace{(e+f)}_{ans2}$$



}

will take less time in this way

↳ Performance via prediction:

predict करुल पावलेन एवंपर pre-ready वाक इला लाकत (मस वरु गायल पाव)।

↳ Hierarchy of memories:



Lec-2

# Yt

# Below your Program:

↳ Application softw: (written in HLL)

↓  
compiler

↳ sys software: machine code

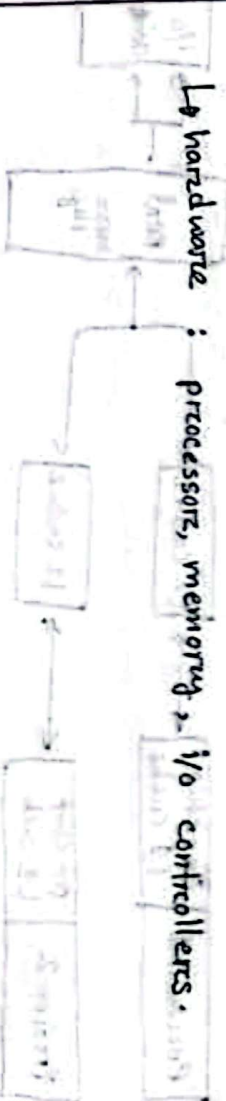
↳ Operating syst: service code

↳ handling i/o

↳ managing memory and storage  
↳ scheduling task

↳ hardware:

processors, memory, i/o controllers.





### \* Inside Processor (CPU)

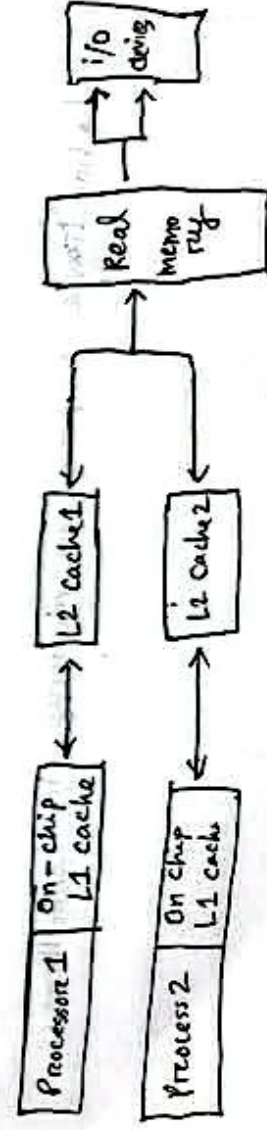
- Datapath: performs operations on data
- Control: sequences datapath, memory
- cache memory: small fast SRAM memory for immediate access to data.

\* ISA: Instruction Set architecture; an abstract interface.

ISA ex MIPS co..

HLL/IR assembly language instruction → convert ~~code~~ → ISA

\* multiprocessor: having two or more processing units (multiple processors) each sharing main memory and peripherals.



\* Response time and Throughput:

↓  
how long it takes to do a task

↓  
total work done per unit time

\* Relative Performance:

$$\text{performance, } p = \frac{1}{\text{execution time, } E}$$

E is the performance of the system

"X is n times faster than Y"

$$\frac{p_x}{p_y} = \frac{E_y}{E_x}$$

Exmpl: ③ time taken to run a program.

— 10s on A, 15s on B

Which time is better? Ans: A

Because,  $p_A$  is faster than  $p_B$ .

$$\frac{p_A}{p_B} = \frac{E_B}{E_A}$$

$$\frac{p_A}{p_B} = \frac{15}{10} = 1.5$$

∴ A is 1.5 times faster than B.

$$\frac{\text{execution time, } E_B}{\text{execution time, } E_A} = \frac{15}{10} = 1.5$$

## \* Measuring Execution time:

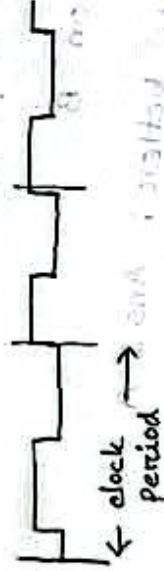
↳ Elapsed time:

↳ CPU Time:

## # CPU clocking:

ଅଟ instruction execute କରିବା ପାଇଁ clock cycle - pass କରିବା ପାଇଁ

↳ clock cycle.



clock period: duration of clock cycle.

ଅଟ cycle ପାଇଁ କାଳ time ନିମ୍ନରେ ।

conversion: ps  $\rightarrow$  ns  $\rightarrow$  s

clock frequency: ଅଟ କରିବା ପାଇଁ cycle ଅଟେ ।

$$\text{Clock freq} = \frac{1}{\text{Clock period}}$$

### # CPU time:

$$\text{CPU time} = \text{CPU clock cycles} \times \text{Clock cycle time/clock period}$$
$$= \frac{\text{CPU clock cycles}}{\text{clock rate}}$$

CPU time: यह दर्शाता है कि कितना समय

clock cycle बसाते जा रहे हैं। CPU time कमाली चाहिए।

performance improved by -

1. Reducing num of clock cycles
2. Increasing clock rate

Exmp:

compA: (2 GHz) clock, 10s CPU time

compB: aim for 6s CPU time.

comp B to require 1/2 times as many clock cycles as comp A for this program.

can do faster clock, but causes  $1/2 \times$  clock cycles

How **fast** must comp B clock be?

rate /  
freq

\* CPU time = Clock cycle  $\times$  clock period

$$* \text{ " } = \frac{C.C}{Cf}$$

given in Comp A, CPU time = 10s

$$C_f = 2 \text{ GHz}$$

Comp B will be, CPU time = 6s

$$\therefore \text{CPU time} = \frac{C_B}{C_f}$$

$$\therefore C_f = \frac{C_B}{\text{CPU time}} \Rightarrow C_f / C_B = \frac{C_B}{6}$$

$$\Rightarrow C_f / C_B = \frac{1.2 \times C_A}{6}$$

$$\Rightarrow C_f / C_B = \frac{1.2 \times \text{CPU time}_A \times C_{fA}}{6} = \frac{1.2 \times 10 \times 2}{6} = 4 \text{ GHz}$$

v-2

env. to 17/2/25

# Instruction Count and [CPI]:

→ Cycles per instruction

$$\text{Clock cycles} = \text{Instruction count} \times \text{Cycles per instruction}$$

→ CPI

$$\text{CPU time} = \text{Instruction count} \times \text{CPI} \times \text{clock cycle time/clock period}$$

$$= \frac{\text{Instr. count} \times \text{CPI}}{\text{Clock rate}}$$

\* CPU time means execution time.

CPI example:

Comp A: Cycle time = 250 ps, CPI = 2.0

Comp B: " = 500 ps, CPI = 1.2

Same ISA.

Which is faster, by how much?

↳

$$\text{CPU Time}_A = \text{Instr Count} \times \text{CPI}_A \times \text{Cycle Time}_A / \text{Clock period}$$

$$= I \times 2.0 \times 250 \text{ ps} = I \times 500 \text{ ps}$$

$$\text{CPU Time}_B = I \times \text{CPI}_B \times \text{Cycle Time}_B$$

$$= I \times 1.2 \times 500 = I \times 600 \text{ ps}$$

$$\frac{\text{CPU Time}_B}{\text{CPU Time}_A} = \frac{600 \text{ ps} \times I_B}{500 \text{ ps} \times I_A} = \frac{I \times 600 \text{ ps}}{I \times 500 \text{ ps}} = 1.2$$

∴ A is faster than B.

# avg CPU time faster, 20% faster.

#CPI more detail:

Instruction & CPI time not necessarily same

ⓐ If different instruction classes take different num of cycles -

$$\text{clock cycles} = \sum_{i=1}^n (\text{CPI}_i \times \text{instruction Count}_i)$$

Weighted avg CPI

$$CPI = \frac{\text{clock cycles}}{\text{Instruction count}}$$

$$= \sum_{i=1}^n \left( CPI_i \times \frac{\text{instr. count}_i}{\text{instr. count}} \right)$$

relative frequency

$$\bullet \text{ CPU Time} = \text{clock cycle} \times \text{clock cycle time}$$

$$= (\text{Instr. Count} \times \text{clocks per Instr.}) \times \text{clock cycle time}$$

Exmp 1:

$$CPI = \frac{24000 \times I}{24000 \times I} = \frac{A}{B}$$

CPI for class

IC in seq 1

[1 batch of code]

IC in seq 2

determine clock cycle?

$$\text{seq-1: } IC = 2 + 1 + 2 = 5$$

$$\therefore \text{clock cycle} = \sum CPI \times IC$$

$$= 1 \times 2 + 2 \times 1 + 3 \times 2 = 10$$

$$\therefore \text{Avg CPI} = 10 / IC = 10 / 5 = 2$$

seq-2:  $IC = 4 + 1 + 1 = 6$

clock:  $\therefore CPI = 1 \times 4 + 2 \times 1 + 3 \times 1 = 9$   
cycle

$\therefore \text{Avg CPI} = 9/6 = 1.5$

\*  $\text{Avg CPI} = \frac{\sum (IC_i \times CPI_i)}{\sum IC_i}$  ;  $IC = \text{Instr. count}$   
 $CPI = \text{clocks per Instr.}$

### Performance Summary:

performance =  $\frac{1}{\text{execution time}}$

cpu time =  $IC \times \text{clock cycle} \times \text{clock cycle time (second)}$

### Power:

power =  $\text{Capacitive load} \times \text{voltage}^2 \times \text{frequency}$

5V ~ 1V [In CMOS IC Technology]

\* Suppose a new cpu has - 85% of capacitive load of old cpu,  
15% v.l 15% freq reduction -

Sol:  $P_{old} = C_{old} \times V_{old}^2 \times f_{old}$

$P_{new} = 0.85 \times C_{old} \times (0.85)^2 \times 0.85 f$

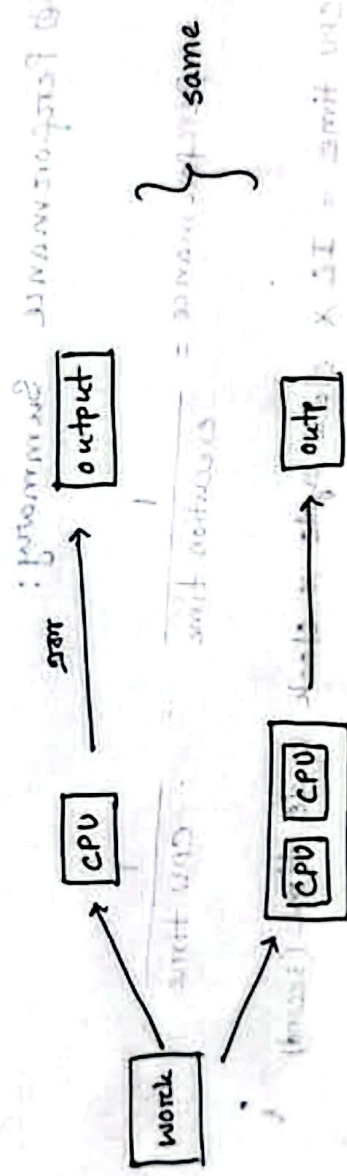
$$\therefore \frac{P_{\text{new}}}{P_{\text{old}}} = 0.52$$

The Power wall : 1) we cannot reduce voltage further

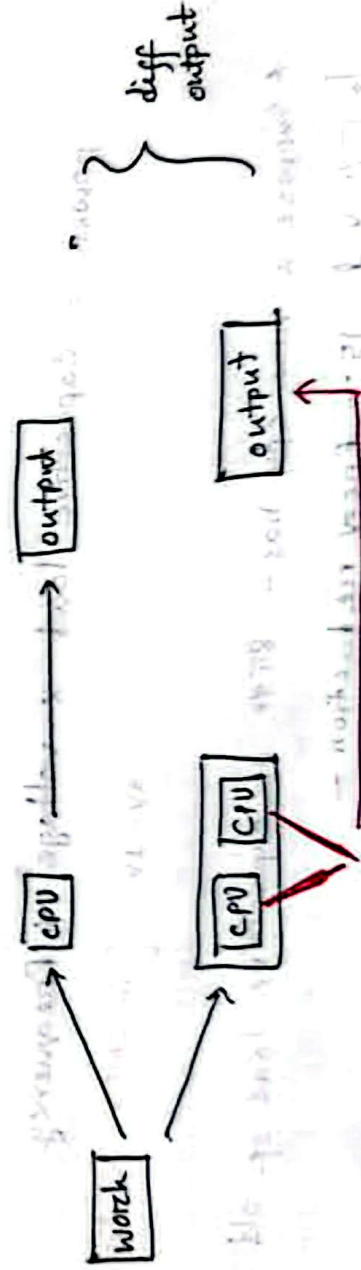
2) we can't remove more heat.

to increase power  $\rightarrow$  start single CPU  $\rightarrow$  start power  $\rightarrow$  power  
 against voltage we can add another.

### Multi-processor :



but parallel programming is very hard,



### Q) SPEC CPU Benchmark:

benchmark value  $\times$  to understand CPU performance.

spec ratio =  $\frac{\text{reference time}}{\text{execution time}}$  → १ बेस लाइन

formula:  $\sqrt[n]{\prod_{i=1}^n \text{Execution time ratio}}$

spec ratio, performance ratio

| Benchmark | IC $\times 10^9$ | CPI  | clock rate (GHz) | Performance |
|-----------|------------------|------|------------------|-------------|
| Geo       | 1970             | 1.10 | 3.2              | 10490       |
| Mean      | 2910             | 0.6  | 3.2              | 9330        |

\* Calculate execution time, spec ratio, summary benchmark (geom mean) overall CPU

execution time =  $\frac{\text{IC (instructions)}}{\text{CPI}} \times \frac{\text{clock time}}{\text{clock period}}$  cycle

[Geo]  $\times \text{CPI}$

$$= \frac{\text{IC} \times \text{CPI}}{\text{clock rate}} = \frac{1970 \times 10^9 \times 1.10}{3.2 \times 10^9}$$

$$= 677.18 \text{ sec}$$

spec ratio =  $\frac{\text{Ref time}}{\text{CPU time}} = \frac{10490}{677.18} = 15.5$  (Ans.)

## determining Summary benchmark

$$\text{CPU time} = \frac{\text{IC} \times \text{CPI}}{\text{Clock rate}}$$

$$= \frac{2910 \times 10^9 \times 0.6}{3.2 \times 10^9} = 545.625 \text{ s}$$

$$\text{spec ratio for 'Hmms'} = \frac{\text{ref time}}{\text{exec time}}$$

$$= \frac{9330}{545.625} = 17.09$$

$\therefore$  Summary/benchmark/

$$\text{geometric mean} = \sqrt[n]{n} \parallel \text{Spec ratio}$$

$$\sqrt[2]{15.5 \times 17.09} = 16.27$$

Ans.

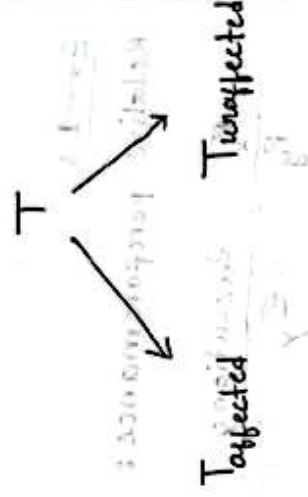
Pitfall: Amdahl's law:

Suppose, 't' is the time to improve the system, optimize the system, and the total time to improve the system is the sum of the time to improve the system and the time to optimize the system.

$$\text{Total time} = \frac{\text{Time to improve}}{\text{Speedup}} + \text{Time to optimize}$$

$$\textcircled{+} \longrightarrow 1 \text{ sec } [T_{\text{affected}}]$$

$$\boxed{- \times \div} \longrightarrow [T_{\text{unaffected}}]$$



$$T_{\text{new}} = \frac{T_{\text{affected}}}{\text{improvement factor}} + T_{\text{unaffected}}$$

$$10 \text{ sec } \textcircled{+} 1 \text{ sec}$$

$$\frac{10}{10} \times \frac{10}{10}$$

$$\frac{10}{10} \div \frac{10}{10}$$

$$\frac{40 \text{ sec}}{40 \text{ sec}} \times \frac{30 \text{ sec}}{30 \text{ sec}}$$

$$\frac{20 \text{ sec}}{20 \text{ sec}} \times \frac{20 \text{ sec}}{20 \text{ sec}}$$

Exmp1: multiply accounts for 80s/100s

• How much improvement in multiply performance to get 5x overall?

$$\frac{100}{5} = 20 \therefore 20 = \frac{80}{n} \times (100 - 80)$$

$$\uparrow 20 = \frac{80 \times 20}{n}$$

$$\therefore n = 80$$

can't be done.

## Assignment

24241083

Sayma Shahid

Ex-1:

Relative performance:

$$\frac{P_x}{P_y} = \frac{\text{Execution}_y}{e_y}$$

$$\Rightarrow \frac{P_x}{P_y} = \frac{15}{10} \Rightarrow \frac{P_x}{P_y} = 1.5 \quad \therefore P_x = 1.5 P_y$$

X is 1.5 times faster than Y.

CPU Time example:

comp-A: 2 GHz Clock, 10s CPU time

$$\text{clock rate}_B = \underline{\hspace{2cm}}$$

$$\text{CPU time}_B = \text{Clock cycles} \times \text{clock cycle} \cdot \text{time/period}$$

$$\Rightarrow \text{CPU time}_B = \frac{\text{clock cycles}}{\text{clock rate}_B}$$

$$\Rightarrow \text{clock rate}_B = \frac{\text{clock cycles}}{\text{CPU time}_B} = \frac{1.2 \times \text{clock cycle}_A}{\text{CPU Time}_B} \quad \text{--- (i)}$$

$$= \frac{1.2 \times \text{CPU Time}_A \times \text{Clock Rate}_A}{\text{CPU Time}_B}$$

$$= \frac{1.2 \times 10 \times 2 \text{ GHz}}{6s} = \frac{1.2 \times 20 \times 10^9}{6s} = \frac{24 \times 10^9}{6} = 4 \text{ GHz}$$

### CPI Example:

Comp A: Cycle time = 250 ps, CPI = 2.0

Comp B: Cycle time = 500 ps, CPI = 1.2

Which is faster?

$$\hookrightarrow \text{CPU time}_A = \text{Instr. Count} \times \text{CPI}_A \times \text{Cycle Time}_A$$

$$= I \times 2.0 \times 250 \text{ ps} = I \times 500 \text{ ps}$$

$$\text{CPU time}_B = I \times \text{CPI}_B \times \text{Cycle Time}_B$$

$$= 1.2 \times I \times 500 \text{ ps} = I \times 600 \text{ ps}$$

$$\frac{\text{CPU time}_B}{\text{CPU time}_A} = \frac{I \times 600 \text{ ps}}{I \times 500 \text{ ps}} = 1.2$$

$$\frac{\text{CPU time}_B}{\text{CPU time}_A} = 1.2 \Rightarrow \text{CPU time}_B = 1.2 \times \text{CPU time}_A$$

$\therefore$  A is faster.

perf. summary:

|         | A      | B      | C      | D      |
|---------|--------|--------|--------|--------|
| 6/7ms   | 500 ps | 500 ps | 500 ps | 500 ps |
| 5/10 ms | 500 ps | 500 ps | 500 ps | 500 ps |
| 5/10 ms | 500 ps | 500 ps | 500 ps | 500 ps |

Q 8:

For wafer X,

$$r = 10 \text{ cm}$$

cost of wafer = 30

$$\text{Total no. of dies} = 200$$

$$\text{Defects per area} = 0.015 \text{ d/cm}^2$$

we know,

$$\text{wafer area} = \pi r^2 = 314.16 \text{ cm}^2$$

Now, Die per wafer = wafer area / Die area

$$\Rightarrow \text{Die area} = \frac{\text{wafer area}}{\text{Die per wafer}} = \frac{314.16}{200}$$

$$= 1.57 \text{ cm}^2$$

For wafer Y,

$$r = 12.5 \text{ cm}$$

cost of wafer = 40

$$\text{Total no. of dies} = 250$$

$$\text{Defect per area} = 0.022 \text{ cm}^2/\text{d}$$

we know,

$$\text{wafer area} = \pi \times (12.5)^2 = 490.87 \text{ cm}^2$$

Now,

Die per wafer = wafer area / Die area

$$\Rightarrow \text{Die area} = \frac{490.87}{250} = 1.96 \text{ cm}$$

21. For wafer X, the num of defective LED dies per wafer  
 $= 314.16 \times 0.015 = 4.71$

For wafer Y, the num of defective LED dies per wafer  
 $= 490.87 \times 0.022 = 10.799$   
 (Ans)

31. Given for X,

Yield = 90%

Die area =  $1.57 \text{ cm}^2$

we know,

$$\text{Yield} = \frac{1}{1 + \left\{ \text{Defect per area} \times \frac{\text{Die area}}{2} \right\}^2}$$

$$\Rightarrow 0.9 \times \left\{ 1 + \left( \text{Defect per area} \times \frac{\text{Die area}}{2} \right)^2 \right\} = 1$$

$$\Rightarrow 1 + \left( \text{Defect per area} \times \frac{\text{Die area}}{2} \right)^2 = \frac{1}{0.9}$$

$$\Rightarrow \text{Defect } \times \text{ area} \times \frac{\text{Die area}}{2} = \frac{1}{3}$$

$$\Rightarrow \text{Defect per area} = \frac{2}{3 \times \text{Die area}}$$

$$= \frac{2}{3 \times 1.57} = 0.42 \text{ defect/cm}^2$$

For wafer Y, Die area =  $1.96 \text{ cm}^2$

$$\text{Yield} = \frac{1}{1 + \left\{ \text{Defect} \times \frac{\text{Die area}}{2} \right\}^2}$$

$$\Rightarrow \text{Defect per area} = \frac{2}{3 \times \text{Die area}} = \frac{2}{3 \times 1.96} = 0.34 \text{ d/cm}^2$$

4) Considering the cost per wafer and the number of defective dies, the cost-effective for high quality LED production is wafer X.

prob 8

1)

|            | A               | B               | C               | D               |
|------------|-----------------|-----------------|-----------------|-----------------|
| PS         | 6               | 3               | 2               | 5               |
| Xbox       | 4               | 4               | 1               | 3               |
| Clock rate | $5 \times 10^5$ | $8 \times 10^5$ | $4 \times 10^5$ | $3 \times 10^5$ |

For PS,

$$\begin{aligned} \text{total clock rate} &= 6 \times 5 \times 10^5 + 3 \times 8 \times 10^5 + 2 \times 4 \times 10^5 + 5 \times 3 \times 10^5 \\ &= 7.7 \times 10^6 \end{aligned}$$

$$\text{Avg CPI} = \frac{7.7 \times 10^6}{2 \times 10^6} = 3.85$$

$$\begin{aligned} \text{For Xbox, total clock rate} &= 4 \times 5 \times 10^5 + 4 \times 8 \times 10^5 + \\ &\quad 4 \times 10^5 + 3 \times 3 \times 10^5 \\ &= 6.5 \times 10^6 \end{aligned}$$

$$\text{Avg CPI} = \frac{6.5 \times 10^6}{2 \times 10^6} = 3.25$$

1) For PS, clock rate =  $4.2 \text{ GHz} = 4.2 \times 10^9 \text{ Hz}$

$$\text{IC} = 2 \times 10^6$$

$$\text{Avg CPI} = 3.85$$

$$\begin{aligned} \text{Execution time} &= \frac{\text{IC} \times \text{CPI}}{\text{clock rate}} = \frac{2 \times 10^6 \times 3.85}{4.2 \times 10^9} \\ &= 1.83 \times 10^{-3} \text{ s} \end{aligned}$$

for X box,

$$\text{clock rate} = 4.8 \text{ GHz} = 4.8 \times 10^9 \text{ Hz}$$

$$\text{IC} = 2 \times 10^6$$

$$\text{CPI} = 3.25$$

$$\text{Exec time} = \frac{\text{IC} \times \text{CPI}}{\text{clock rate}} = \frac{2 \times 10^6 \times 3.25}{4.8 \times 10^9} = 1.35 \times 10^{-3} \text{ s}$$

for HPC,

$$\text{clock rate} = 5 \text{ GHz} = 5 \times 10^9 \text{ Hz}$$

$$\text{CPI} = 1.8$$

$$\text{IC} = 2 \times 10^6$$

$$\text{exec time} = \frac{\text{IC} \times \text{CPI}}{\text{clock rate}} = \frac{2 \times 10^6 \times 1.8}{5 \times 10^9} = 7.2 \times 10^{-4} \text{ s}$$

$\therefore$  HPC has better performance than PS & X box.

3) Given,

$$\text{Exec. Time of PS} = 1.83 \times 10^{-3} \text{ s}$$

$$,, \text{ X box} = 1.35 \times 10^{-3} \text{ s}$$

$$,, \text{ PC} = 7.2 \times 10^{-4} \text{ s}$$

$$\text{Now, spec ratio} = \frac{0.1}{1.83 \times 10^{-3}} = 54.64 \quad (\text{PS6})$$

$$\text{Again, spec ratio (X box)} = \frac{0.1}{1.35 \times 10^{-3}} = 74.07$$

$$\text{spec ratio (PC)} = \frac{0.1}{7.2 \times 10^{-4}} = 138.9$$

4) For P56,

$$\text{Exec. time} = 1.83 \times 10^{-3} \text{ s}$$

$$\text{Avg CPI} = 3.85$$

$$\begin{aligned}\text{Reduced exec. time} &= \left( \frac{35}{100} \right) \times 1.83 \times 10^{-3} \\ &= 1.83 \times 10^{-3} \cdot - 6.405 \times 10^{-4} \\ &= 1.1895 \times 10^{-3} \text{ s}\end{aligned}$$

$$\text{Increased CPI} = \frac{25}{100} \times 3.85$$

$$= 3.85 - 6.96 = 2.89$$

$$\text{we know, exec. time} = \frac{\text{IC} \times \text{CPI}}{\text{clock rate}}$$

$$\Rightarrow \text{clock rate} = \frac{2 \times 10^6 \times 2.89}{1.1895 \times 10^{-3}} = 4.859 \times 10^9 \text{ Hz}$$

5) for Xbox,

$$\text{clock cycle of B} = 3.2 \times 10^6 \times \frac{30}{100}$$

$$\begin{aligned}\text{Reduced B} &= 3.2 \times 10^6 - 9.6 \times 10^5 \\ &= 2.24 \times 10^6\end{aligned}$$

$$\text{clock cycle D} = .3 \times 9 \times 10^5 = 2.7 \times 10^5$$

$$\text{Reduced D} = 9 \times 10^5 - 2.7 \times 10^5 = 8.97 \times 10^5$$

$$\begin{aligned}\text{Avg CPI} &= 2 \times 10^6 + 2.24 \times 10^6 + 4 \times 10^5 + 8.97 \times 10^5 \\ &= 5.537 \times 10^6 / 2 \times 10^6 = 2.7685\end{aligned}$$

$$\text{clock rate} = 4.8 \times 10^9 \text{ Hz} \times \frac{10}{100} = 4.8 \times 10^8$$

$$= 4.8 \times 10^9 - 4.80 \times 10^6 = 4.32 \times 10^9 \text{ Hz}$$

we know, Exec time =  $\frac{IC \times CPI}{\text{clock rate}}$

$$= \frac{2 \times 10^6 \times 2.7685}{4.32 \times 10^9} = 1.28 \times 10^{-3} \text{ s}$$

No, it still won't outperform the HPC.

Prac. Prob:

$$1) \text{ speed factor} = \frac{1}{(1-p) + \frac{p}{S}} = \frac{1}{(1-0.4) + \frac{0.4}{1.67}} = \frac{1}{0.6 + 0.24} = 1.19$$

2) Here, New time of frame decoding =  $200 \times \frac{350}{500} = 140 \text{ sec}$

$$\text{Speed factor} = \frac{200}{140} = 1.43$$

$$\text{Time for color correction} = 150 \times \frac{350}{500} = 105 \text{ sec}$$

$$\text{Speed factor} = \frac{150}{105} = 1.43$$

$$\text{Time for edge detection} = 100 \times \frac{350}{500} = 70 \text{ s}$$

$$\text{speed factor} = \frac{100}{70} = 1.43$$

3) Here, execution time = frame decoding + edge detection  
 $= 200 + 100 = 300 \text{ s}$